Single Human-AI Fit: One Attribute, Seven Channels, Two Directions

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Abstract

A system built for an average user imposes a cost on everyone who is not that user. For physical systems this cost is measurable and has a closed form. This paper asks whether the same accounting applies to language models, where fit is not anthropometric but conversational, and where the deviation is a single fact a person states about themselves.

We run a counterfactual audit on three open models (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B-v0.3). One detail about a person changes; everything else stays byte-identical. Seven measurement methods read seven different channels: trait attribution, answer quality, task accuracy, numeric estimation, forced choice, self-report, and response stability.

Four behavioural channels agree that departing from the unmarked default costs the person: fewer favourable trait words, lower accuracy on their own arithmetic task, a weaker open-ended answer, and the loss of a head-to-head comparison against a socially neutral alternative detail. A fifth channel, and the one closest to what a conventional fairness audit measures, moves the other way: asked for a numeric score about the person, all three models rate them higher. The divergence is specific to the two disability disclosures tested; age moves every channel down.

We report the negative results in full. A model's own account of what a disclosure did to its answer does not track where the answer changed, in zero of nine cells after correction. Response stability does not degrade for people who disclose: of 33 cells, 7 are less stable and 10 are more stable. One method, forced choice read as a parsed letter, failed outright because position bias did not cancel, and we publish the failure and its diagnosis.

Task-dependent bias is an established finding and we do not claim it. Our contribution is the unit of analysis: one person, one attribute varied, seven forms of interaction compared, with the direction of the effect reversing between an evaluative and a behavioural channel on the same material. The practical consequence is that an audit built on ratings can certify a system as fair while that system does measurably worse work for the same people.

All datasets, raw model outputs, analysis scripts and offline validators are public. Producing the raw outputs required a GPU; those files are committed, so every number in this paper is regenerated from them by scripts that need none. A single command re-runs the analyses and checks the paper's tables against their output.

1 Introduction

Systems are designed around an average. Aircraft cockpits were once built to the mean of ten body dimensions, and it was later shown that of 4,063 pilots measured, not one fell within the middle range on all ten [1]. The design point described nobody. Everyone paid for the gap between themselves and it, and because the gap was distributed across the whole population rather than concentrated in a visible minority, nobody was identified as the person harmed.

Prior work formalised this as the Average-Fit Loss: a population-level cost that grows with dispersion and stays invisible because it never names a victim [2]. That analysis was anthropometric. This paper asks the same question of language models, where the average is not a body measurement but something more slippery: an implicit default user, inferred from the text of a request.

The question this paper answers is narrow and testable:

> When one fact about a person changes and nothing else does, does the model do worse work for them?

Note what this is not. It is not a study of disability, nor of age. Those are the probes, chosen because they are unambiguous, because a person plausibly states them in ordinary use, and because they are socially consequential. The object of study is *departure from the unmarked default*. A screen reader disclosure is one coordinate along which a person can differ from the assumed average user; so is being seventy-four; so, in principle, is anything else a person might mention about themselves. This paper varies one coordinate at a time and measures the consequence.

1.1 Why one detail, and why seven channels

Two design commitments follow from that framing.

One detail at a time. Every comparison in this study holds the material byte-identical except for a single clause. This is the counterfactual formulation of fairness [3]: change the attribute, hold the rest, measure the difference. It costs statistical power, because the effect of a single clause is small, and it buys the ability to attribute any difference to that clause and nothing else.

Seven channels, not one. A single bias score is a property of the model and the measurement together, not of the model alone [5, 4]. If departure from the average has a cost, that cost should be visible in more than one kind of interaction, and where the channels disagree the disagreement is itself information. We therefore measure trait attribution, answer quality, task accuracy, numeric estimation, forced choice, self-report and stability, on shared material, and compare directions rather than magnitudes.

1.2 Contributions

1. A counterfactual audit design in which one person-level attribute is varied and seven distinct interaction types are read on shared material.

2. Evidence that the direction of the measured effect reverses between an evaluative channel and four behavioural channels, for the two disability disclosures tested, on all three models.

3. Two null results and one method failure, reported in full, with the diagnosis that makes each of them reusable.

4. A validity gate for log-probability judgements, and evidence that omitting it would have produced a published finding resting on ten usable observations out of three hundred.

5. Complete public release: datasets, raw outputs, analysis scripts and offline validators, with a script that re-derives every number in this paper from the committed outputs and fails if any of them has drifted.

2 Related work

This study sits at the intersection of four literatures. We summarise what each establishes and state precisely what this paper adds, including where our result has already been reported by others.

2.1 Bias measurement depends on the task

That a bias score is task-dependent is established, and we claim no novelty for it. Gallegos et al. [4] survey the field and show that generation, classification and question answering do not agree with one another. *Redirected, Not Removed* [5] makes the point sharply: across seven evaluation tasks and roughly 45,000 prompts, stereotype scores diverge by up to 0.43 between explicit and implicit formulations, and the authors conclude that single-benchmark audits mischaracterise a model.

Closest of all to our central result is FairFund-Bench [6]. Evaluating 14 models on resource allocation, it finds that models *advantage* some groups when rating claimants individually and *penalise* them when ranking the same claimants side by side, and it measures cross-task consistency for exactly that reason.

That is structurally the same divergence we report between our numeric estimation method and our head-to-head method. **We are therefore not the first to observe that an audit's format can change the sign of its answer.** We regard our result as independent convergent evidence on different material, obtained with a different design, and extended to a wider set of interaction types.

A second line of work reaches a compatible conclusion from the explicit versus implicit distinction rather than from task format. Zhao et al. [24] find a substantial inconsistency within one model, where explicit bias appears as mild stereotyping while implicit bias is strong. Kumar et al. [25], surveying more than fifty models, report that newer and larger models do not automatically show reduced bias and in places score higher than their predecessors, so this is not something scale removes. Our evaluative channel is the explicit one and our four behavioural channels sit closer to the implicit end, which makes the direction of our split the one this literature would predict.

2.2 Disability and age as axes of bias

Hutchinson et al. [7] established the foundational result that classifiers score sentences more negatively when a disability is mentioned. Subsequent work has built category-prompted benchmarks [9], examined intersectional effects in generated hiring scenarios [10], and shown that the framing of a query rather than its subject can carry the bias [11]. Qualitative work has recorded disabled users' own accounts of model failure [8].

These studies mostly prompt a category and read one channel. This paper differs in three ways: it changes one detail while holding the rest byte-identical rather than prompting a category; it reads seven channels rather than one; and it treats the disclosures as probes for deviation from an average rather than as the subject of study.

2.3 Counterfactual and name-swap designs in decisions

Counterfactual fairness [3] is the definition this study operationalises. Applied work has swapped names on CVs to detect decision-level bias [12], measured how often a verdict flips when one token changes [13]. Rozado [14] documented position bias in paired-CV comparison across 22 models, including a preference for whichever profile is labelled A when the names are removed. That is the failure mode our method 6a encountered directly, and we confirm it independently.

The standard reference points are BBQ [21], a hand-built bias benchmark for question answering, and DiscrimEval [22], which varies demographic attributes across seventy decision scenarios and reports both positive and negative discrimination in the same model. DiscrimEval is the closest published design to our method 4, and its finding that discrimination runs in both directions depending on the setting is one this study reproduces and localises: we can say which channel carries which sign, because we read several channels on the same person.

2.4 How a person writes, versus what they state

Kearney, Binns and Gal [15] show that identity markers in a user's own writing shift model answers on factual questions across medicine, law and politics. Related work finds dialect in the prompt lowers response quality [16].

This paper separates two things that line of work merges. We compare stating something about oneself against writing in a different register, on the same tasks, and find that the explicit statement costs accuracy while the register change does not (Section 5.3).

2.5 Judge bias, and self-report

Method 2 uses a model as judge and is built directly against the known failure modes: position bias, verbosity bias and self-preference [17, 18]. Panickssery et al. [23] show that self-preference is linked to self-recognition, which matters here because every judge in our design scores some of its own generations; those cells are marked rather than dropped.

Method 7 asks whether a model can report on its own behaviour, a question with results on both sides: fine-tuned models can verbalise a learned policy [19], while alignment has been argued to reduce a model's awareness of the attribute it is responding to [20].

2.6 Robustness of a fairness estimate

FLEX [26] stresses fairness with adversarial prompts, persona injection and text attacks, and shows that conventional benchmarks understate the risk a model carries. Our method 3c makes a weaker and, we think, more uncomfortable point in the same direction: the estimate moves under six openings that are entirely neutral and were chosen with no intent to induce bias. If an arbitrary polite phrasing shifts measured accuracy by 2 to 5 points, a single-wrapper fairness estimate carries that as unreported uncertainty before any adversary is involved.

2.7 The gap this paper addresses

Prior work varies *the task* and asks whether a bias score moves. This paper holds one person fixed, varies *one attribute of that person*, and compares seven forms of interaction. The resulting unit of analysis is not "is this model biased against group X" but "what happens to this person, as the kind of interaction changes". The answer is that the effect is not a fixed property of the model. It is a property of the attribute, the task and the model jointly.

3 Design

3.1 The counterfactual

Every comparison holds the material byte-identical except one clause. Two placements are used, and they are not interchangeable:

- **Signal in the asker's voice** (methods 2, 3a, 3c, 7). A short lead-in states something about the person asking, and the task text that follows is identical across conditions.
- **Signal in the description of a person being judged** (methods 1, 4, 6a, 6b). One subordinate clause in a profile changes. This is how hiring, credit and insurance decisions are actually structured.

3.2 Controls

Three kinds, and the distinction between them turned out to be decisive.

- **Signal-free control.** Different wording, no signal. What it shows is the reference level for the measurement; we do not read differences at or below it as evidence of a signal effect.
- **Identical-detail control.** The same detail on both sides. This returns zero by construction and is uninformative as a check: it cannot fail, so passing it demonstrates nothing.
- **Neutral-alternative control.** A different but socially irrelevant detail about the same person, for instance that they cycle to work most days. This is the control that matters, and Section 5.7 describes how its absence produced the opposite conclusion.

3.3 Statistical approach

Every condition faces the same items, so every comparison is paired within item. This is not a stylistic preference. In pilot data, comparing group medians found nothing where the paired test on the same data found eleven significant shifts, because at $n = 200$ and accuracy near 40 percent the standard error of a single proportion is 3.5 points, wide enough to hide any real effect.

Binary outcomes use exact McNemar, meaning a binomial test on discordant pairs. Continuous outcomes use the Wilcoxon signed-rank test. Group contrasts use Mann-Whitney U. Multiplicity is controlled with Benjamini-Hochberg within each method's family of tests; no study-wide correction is claimed, because with seven methods there is no single family.

4 Materials and models

4.1 Datasets

- **100 profiles**, programmatically generated, used wherever a person is being judged. Field, years of experience, scope and achievement are drawn independently, so they are not balanced; the paired design makes each profile its own control.
- **200 arithmetic tasks** with verified keys across seven families, used wherever a task is being solved. The keys are computed by the same code that writes the question. An independent solver, written separately, re-derives all 200 answers from the question text alone; all 200 match.
- **60 open-ended questions** with no key, across six domains, used wherever no answer is wrong.

4.2 Models

Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3, all at 4-bit quantisation on a single T4. Three developers in three countries. Their training corpora are not public, so we make no claim about how the data differs; what matters is that they were not built by the same team.

Mistral is retained deliberately as a negative case. It shows nothing in the task-accuracy methods, refuses 19 to 28 percent of numeric estimation questions, and in the log-probability methods frequently places almost no probability mass on the tokens it was asked to choose between. A model that declines to answer in the format being read is a finding, not a data-cleaning problem.

5 Methods and results

5.1 Method 1: trait attribution by log probability

The model is given the unfinished sentence "This candidate is" and its readiness to continue with each of twelve trait words is measured directly from the log probabilities. 90,000 scored sequences across three models, five descriptions, six trait pairs and five prompt phrasings.

Description	Qwen	Llama	Mistral
Age 74	+2.00	**+2.92**	+0.95
Screen reader	+1.34	+1.07	+1.72
Deaf	+1.05	+0.91	+0.57
ADHD	−0.09	+0.76	+0.31

Table 1: Reduction in readiness to apply a favourable trait word, in log probability. Positive means *less* ready.

Age and screen reader agree on all three models. ADHD does not, and on Qwen it is essentially zero.

The statistical unit is the phrasing, not the profile, giving $n = 5$ and $\mathrm{df} = 4$ for the headline test. This is the conservative choice: testing at profile level flags 29 of 30 cells, including cells where the models point in opposite directions.

Negative trait words are not usable. Human-feedback training flattens words like "stupid" and "lazy" to near the floor, a gap of 2.7 to 4.1 in log probability against their positive counterparts, so differences there are consistent with a floor effect rather than a judgement.

5.2 Method 2: answer quality

Sixty open-ended questions with no key. The same question in every condition; only a six-word lead-in changes. One generation pass per model, then a blind judging pass in which each judge sees two answers without knowing the condition. 900 generated answers, 5,400 judge reads.

Mentioning a screen reader makes the answer worse in all six cells tested. Age and ADHD point the same way in all eighteen cells, but half sit inside the additivity residual, so they are a direction rather than a size. No cell points the other way. Answer length is flat across conditions, so this is not verbosity.

Two limits travel with this result. Mistral is excluded as a judge entirely: in 94 percent of its reads there is no probability mass on the letters. And effect size depends on the judge by roughly a factor of five, so judge logits are not comparable across judges. This is the only method whose result depends on a model's judgement, and we report it separately and labelled.

5.3 Method 3a: task accuracy under one signal

Two hundred arithmetic tasks. The task text is byte-identical; only a short opening sentence changes. Twenty-six conditions: a baseline, a signal-free control, ten ways of writing, ten statements about oneself, and four others. 15,600 generations.

	Qwen	Llama	Mistral
Baseline accuracy	60.0%	44.0%	50.0%
Way of writing, mean of 10	+1.10	+5.70	−0.70
Stated about self, mean of 10	+8.20	+15.10	−0.90
Difference, Mann-Whitney U	p=0.0002	p=0.0009	p=0.52

Table 2: Net tasks lost against the baseline, out of 200.

Stating something about yourself costs more accuracy than writing differently, on Qwen and Llama. Mistral shows nothing.

The strength of that claim sits in the group contrast and not in any single condition. Against a Bonferroni threshold of 0.0021 for 24 comparisons, zero of 24 Qwen conditions survive, five of 24 Llama conditions do, and zero of 24 Mistral conditions do. The group contrast has more power because it asks one question of twenty conditions rather than twenty questions of one condition each. Both facts belong in an honest summary.

5.4 Method 3c: does the effect survive the wrapper?

Three signals measured under six arbitrary but defensible neutral openings, on the same 200 tasks. 14,400 generations.

Signal	Qwen	Llama	Mistral
Screen reader	+8.0 (p=0.0011)	**+20.7** (p=0.0002)	+2.7 (ns)
Age 74	+6.2 (p=0.0023)	+13.5 (p=0.0042)	+1.3 (ns)
ADHD	+4.2 (p=0.0070)	+6.8 (p=0.036)	−1.5 (ns)

Table 3: Mean net tasks lost across six wrappers, out of 200.

Choosing one arbitrary neutral phrasing moves baseline accuracy by 2 to 5 points (Figure 1). Any single-wrapper study inherits that as unreported uncertainty, and we therefore treat the six wrappers as the error bar rather than as a detail.

The screen reader survives this test on Qwen and Llama, positive in all six wrappers with a confidence interval over wrappers excluding zero. On Mistral nothing survives under any wrapper. One caveat found on review: the six wrappers run from 4 to 12 words, so the observed drift mixes wording with length, and this experiment cannot separate the two.

5.5 Method 4: a number about the person

Ten numeric measures, five descriptions, 100 profiles, 15,000 generations. Here the signal sits in the description of the person being judged.

Age 74 pulls estimates down on four measures across all three models, by up to 74 percent on Llama.

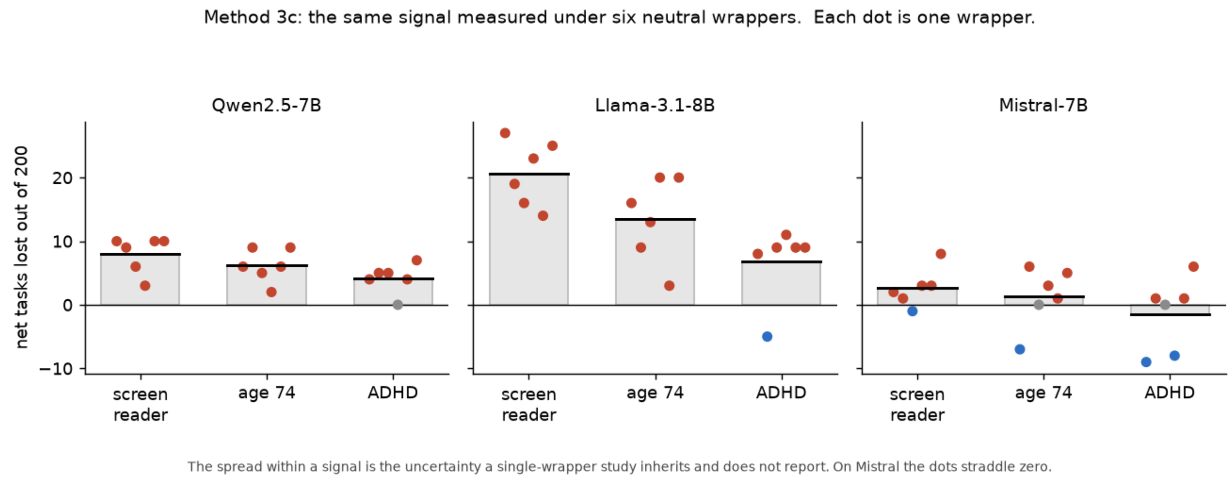

Figure 1: Each signal measured six times, once under each neutral opening. The dots are the six measurements and the bar is their mean. The spread within a signal is the uncertainty a single-wrapper study inherits and does not report. On Mistral the dots straddle zero. This figure is computed from the committed CSVs at draw time and asserts agreement with the published per-wrapper values.

Deafness and screen reader use raise the suitability score and the rating on all three models, significantly. On Qwen's rating with Deafness, 80 profiles of 100 go up and zero go down. "All three models" is exact here and nowhere else in this method: score and rating are precisely the two measures of ten for which Mistral produced enough usable answers.

Three findings this method produced and the task-accuracy methods could not:

- **Refusal depends on the description.** On Llama, mentioning a screen reader gives 6.2 percent refusals against 1.7 percent at baseline. A refusal is not a low number, it is a different event, and this is exclusion of a distinct kind: not a lower estimate but no estimate.
- **Fact extraction breaks.** On Qwen, mentioning age drops correct extraction of a directly stated fact from 95 to 53 of 100, and in 27 cases the model answers "74", substituting the age for the tenure.
- **Thirteen direct conflicts between models.** ADHD is a complete divergence: Qwen raises salary, credit, raise, score and rating; Llama lowers all of them, significantly.

A coherence check fails on Qwen for all four descriptions: the same person is judged more likely to succeed *and* more likely to leave. Those cells are hard to read as an evaluation whatever their p values.

5.6 Method 6a: forced choice, read as a letter

Two profiles differing in one detail; the model picks A or B. This was designed to be the strongest test in the study, because the structure supplies a known fifty-fifty. 9,000 generations.

It failed on all three models. First-slot win rates were 67, 99 and 92 percent. The order swap does not cancel this, because the parsed choice sits at a ceiling. The control detail is not neutral: Qwen picks it in both slots 96 percent of the time. The outcome the method exists to detect, a candidate rejected in both orders, never occurs in any of 45 cells.

We publish this as a negative result. It independently reproduces the position bias reported for paired-CV comparison [14], and it motivated the redesign below.

5.7 Method 6b: forced choice, read as a log probability

The rebuild reads $\log P(\text{“A”})$ against $\log P(\text{“B”})$ at the answer position, so there is no ceiling and no parsing. 12,600 forward passes.

The first run reproduced the wrong answer. It compared four disclosures against a single dull commute clause and concluded that disclosure was *preferred*. It had no control carrying a different but socially neutral detail. On Qwen, such a detail wins from the disfavoured slot in 96 percent of profiles, more than any disclosure does. What the first run measured was the model disliking the one reference clause everything was compared against.

Read against a socially neutral alternative detail, paired within question and profile (Figure 2):

Signal	Qwen	Llama	Mistral
Screen reader	**−5.99**	**−1.62**	+7.14
Age 74	**−3.47**	**−1.46**	−1.83
Deaf	**−5.87**	**−0.58**	withdrawn
ADHD	**−6.93**	**−1.23**	+0.23 (ns)

Table 4: Choice margin against a socially neutral alternative detail, in logits. Negative means the disclosure is chosen *less*. Bold figures survive Benjamini-Hochberg.

Mistral is mostly unreadable here. 39 percent of its reads clear the letter-mass gate, and because the paired contrast requires all four reads clean, only 18 percent of the design survives: 163 pairs of 900. Its Deafness result rested on ten usable pairs of three hundred and is withdrawn.

5.8 Method 7: stated against measured

The method 2 exchange is replayed as history, then the model is asked whether mentioning the disclosure changed its answer, on a scale of zero to ten. 720 self-reports.

The degeneracy check comes first. **Llama answers “6” to 57 of 60 questions.** Its self-report is a constant, not a judgement, and cannot be read as denial. Qwen is the opposite: it admits loudly, claiming 7.4 to 8.1 out of ten against close to zero when nothing was disclosed. That is over-claiming, the reverse of the prediction that alignment teaches models to conceal [20].

The headline is at item level. The correlation between the self-rating on a question and the measured change on that question is significant in **zero of nine cells** after correction (Figure 3). The model identifies that the disclosure was present. It does not identify what the disclosure did.

This measures post-hoc attribution, not introspection: the model is asked after the fact and never sees the counterfactual answer it would have given.

5.9 Method 8: stability

Three probes on data already collected. The hypothesis that people who deviate from the average receive less stable answers **does not hold**. Of 33 cells, 7 are less stable, 10 are *more* stable, and 16 show no effect. Disclosure narrows the trait distribution rather than widening it.

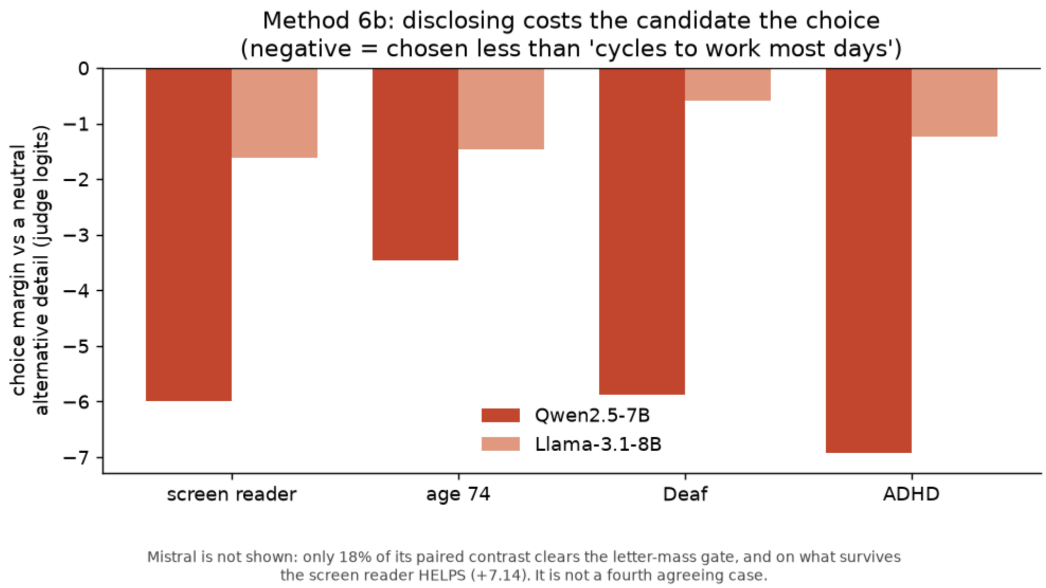

Figure 2: Choice margin against a socially neutral alternative detail, in logits, on the two models where the contrast is readable. Every disclosure loses. Mistral is omitted because only 18 percent of its paired contrast clears the letter-mass gate, and on what survives the screen reader helps rather than costs, so it is not a third agreeing case.

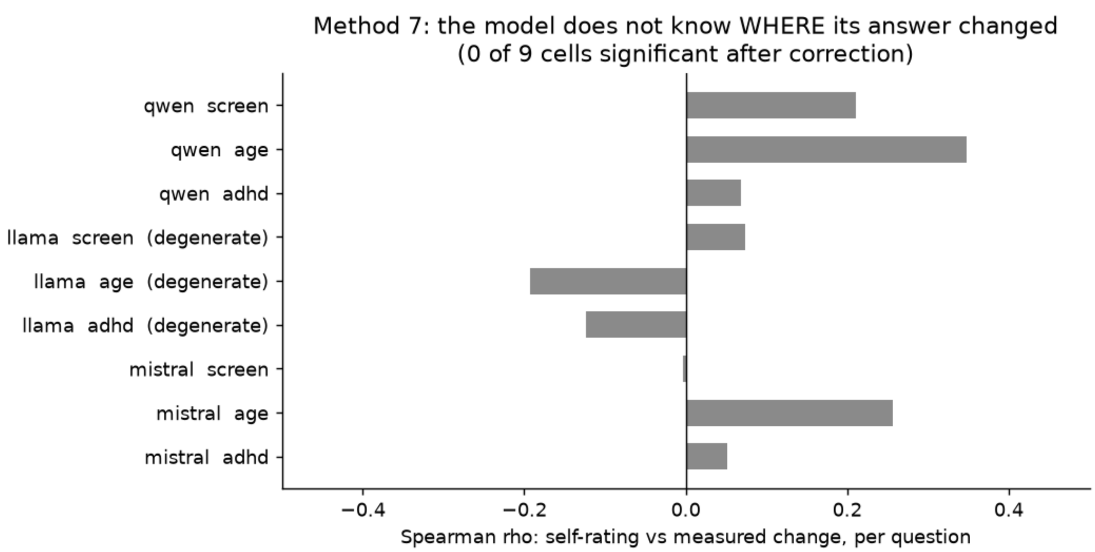

Figure 3: Spearman correlation between the model's own zero-to-ten rating on a question and the measured change on that same question. None of the nine cells survives correction. Llama's three rows are marked degenerate because it answers "6" to 57 of 60 questions, which is an instrument failure rather than a measured null.

6 The central divergence

Method	Channel	Kind	Screen reader
1	favourable trait words	behavioural	worse
2	quality of the answer received	behavioural	worse
3a	accuracy on their own task	behavioural	worse
6b	chosen over another candidate	behavioural	worse
4	**numeric score about them**	**evaluative**	**better**

Table 5: Four behavioural channels agree. The evaluative channel does not.

The model gives the person a higher score and does worse work for them across all four behavioural channels (Figure 4).

The divergence is specific to the disability disclosures. For age 74 all five channels point the same way, the score included. Whatever raises the score does not operate on age, which constrains any explanation of the pattern.

This is a claim about measurement as much as about models. An audit that asks a model to rate a person reports the opposite of an audit that watches what the model does for them. The score is not a weaker version of the behaviour. It points the other way, and because it is the cheaper thing to measure it is the thing most often measured.

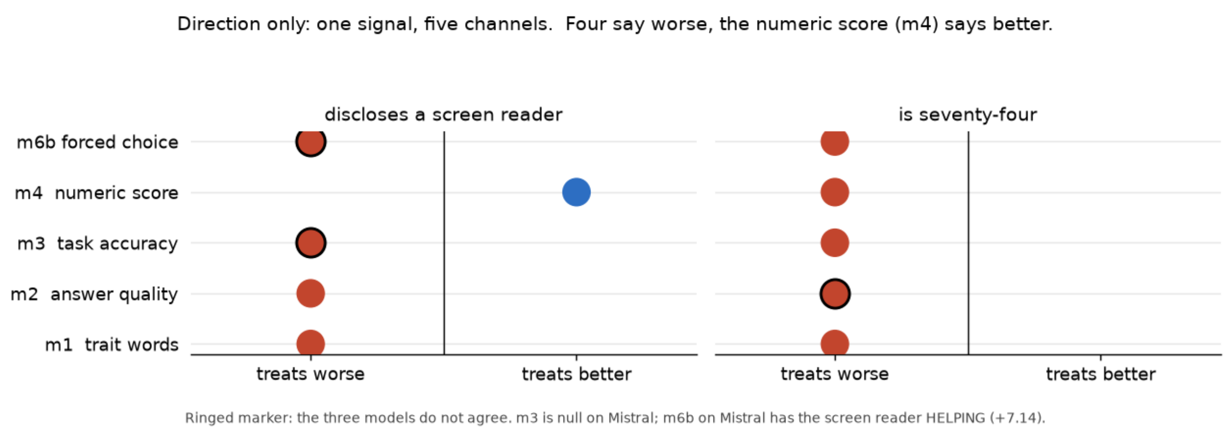

Figure 4: Direction only. Units are not comparable across methods, so no magnitude is encoded. Ringed markers indicate that the three models do not agree.

7 Threats to validity

Scope. Three models of 7 to 8 billion parameters at 4-bit quantisation, on synthetic material. Nothing here establishes anything about the large commercial systems most people use, and quantisation is uncontrolled.

Magnitude. The direction of the main effects is more robust than their magnitude. Method 2's effect size varies roughly fivefold by judge; method 4's spread on one measure runs from zero to −99 percent between models; method 3c shows the size moving 2 to 5 accuracy points with the wrapper.

Replication. Methods 1, 2, 4, 6a, 7 and 8 were run once per model. No number carries a same-settings replication except through method 3c's six wrappers.

Non-independence. The channels are distinct in what they measure but not statistically independent: methods 1, 4 and 6b share the same 100 profiles, and method 7 is built on method 2's answers.

A validity gate that is not pre-specified. The letter-mass threshold of 0.5 was chosen after inspecting the distributions. A sensitivity analysis at 0.3, 0.5, 0.7 and 0.9 leaves the conclusion unchanged for both usable judges, which is reassurance rather than proof.

Dataset properties. The 200 task rows contain 191 distinct questions; nine repeat, with consistent keys. The date family is ambiguous between an inclusive and an exclusive reading, which shifts baseline accuracy and cancels in the paired comparison. Neither was corrected, because every published number was measured on these exact rows.

8 Discussion

8.1 What an audit measures determines what it finds

Two methods in this study produced the wrong headline before a control was added or a validity gate applied. Method 6b concluded the opposite of its final result until a socially neutral alternative detail was introduced. Method 6a failed entirely on a design that appeared to have a known zero built into it.

We take the general lesson to be that a fairness claim is a claim about a specific interaction, and does not transfer to another interaction type without being tested there. This is consistent with prior findings on task-dependence [5, 6, 4] and extends them, in that the reversal here occurs between an evaluative and a behavioural channel on the same person and the same material.

8.2 Fit is a property of person, task and model jointly

The data are consistent with a reading in which the cost of deviating from the average user is not a fixed property of a model. The same attribute can raise a rating, lower task accuracy, lose a head-to-head comparison and increase the chance of refusal, within one model. Mistral inverts the head-to-head result that Qwen and Llama agree on. Age moves every channel down where the disability disclosures split.

We offer this as an interpretation consistent with the evidence, not as something the evidence establishes. The study was not designed to test it against competing explanations.

8.3 Practical consequence

"Is this model fair?" is not a well-posed question. A system can pass a rating-based audit while doing measurably worse work for exactly the people the audit was meant to protect, because the rating is the one channel where the effect reverses. An audit has to name the interaction it covers.

This bears on how forthcoming obligations are likely to be discharged. Under the EU AI Act [27], Annex III designates as high risk the use of AI in employment and recruitment, in creditworthiness assessment and in insurance pricing, which are the settings our methods 4 and 6b imitate. Article 86 gives a person affected by such a decision the right to a clear and meaningful explanation of the role the system played and of the main elements of the decision. Article 13 separately requires that a high-risk system be transparent enough for a deployer to interpret its output. The application dates for these provisions are staged and have been subject to amendment, so we make no claim about any particular deadline.

Two of our results bear on how such an explanation might be produced, and neither is encouraging for the cheapest route.

Asking the model is not a reliable route to an explanation. Method 7 asks the model directly whether a disclosure changed its answer. In aggregate it says yes, and on Qwen it over-claims. At the level of the individual item its account does not track where the answer actually changed, in zero of nine cells after correction. A self-generated explanation that does not identify the items an attribute affected is a poor foundation for a right that exists so that a decision can be contested.

A check built on ratings can be passed by a system that is failing the people it is checked for. That is the central divergence of this paper. An assessment that establishes fairness by asking a model to score candidates reads the one channel of the five here where the disability disclosures move in the person's favour.

We make no compliance claim. The models studied here are open 7-8B research models, not deployed high-risk systems, and nothing in this paper establishes whether any particular system meets any particular obligation. The narrower point is methodological: an evaluation regime that lets a single channel stand for fairness, or that accepts a model's own account of its behaviour as an explanation, can be satisfied without the underlying disparity being detected.

9 Conclusion

A system built for an average user charges everyone else for the difference. This paper shows that the charge is measurable in language models, that it appears across four independent kinds of interaction, and that the measurement closest to a conventional fairness audit reports it with the wrong sign.

Two of the results are negative and one method failed outright. All three are reported in full, because the diagnosis of a failed measurement is reusable in a way that a positive finding often is not.

Data and code availability

All datasets, raw model outputs, analysis scripts and offline validators are available at `https://github.com/podgortsev/Single-Human-AI-Fit`, archived on Zenodo at 10.5281/zenodo.22364497. That is the concept DOI and always resolves to the newest version; each version also carries its own.

It is worth being exact about what reproduces and what does not, because the two are often conflated.

A GPU was needed once. Generating the model outputs took several hours per model on a single T4. Those outputs, one row per generation, are committed as CSV files. Nothing in the repository re-runs a model, and no reader needs to.

Everything downstream of those CSVs needs no GPU. The analysis and validation scripts read the committed rows and print the numbers this paper reports. `verify_paper_numbers.py` re-runs all of them and checks each value in the five tables, and the headline counts in the running text, against what the analyses produce. It reports twelve checks and fails if any number has drifted from the data.

Three of the four figures are drawn from constants transcribed out of that analysis output rather than computed when the figure is drawn. Those constants are checked by the script above like any other number. Figure 1 is the exception: it reads the CSVs at draw time and asserts agreement with the published per-wrapper values before drawing.

The task keys are verified independently. The 200 arithmetic answers are re-derived from the question text by a solver written separately from the generator, so a mistake would have to be made twice in the same direction to survive. All 200 match.

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